Q10: Comprehension Check

DLS 2026 — Kukla

October 2026

Comprehension check. A strong model correctly derives a calculus result but says the word "banana" has two a's. Give the most likely technical reason for the counting error, and name a change to the *input or harness* (not the model weights) that would fix it.

Answer. The model does not see individual letters—it processes sub-word *tokens*, so "banana" is one or a few tokens with no explicit per-letter inventory, and counting a specific letter requires structure the representation does not expose. Fixes that leave the weights untouched: reframe the input so the letters are individually visible (e.g. ask it to first spell the word out letter-by-letter, "b-a-n-a-n-a", then count), or give it a tool (run a tiny code snippet / character counter). Either exposes the sub-token structure the raw tokenization hides. ("banana" in fact has three a's—exactly the kind of slip the reframing prevents.)